

COMPARATIVE ANALYSIS OF BASIC DENOISING METHODS FOR SPEECH IMPROVEMENT USING STATISTICAL MEASURES

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Abstract

This research presents an exploratory analysis on the effectiveness of four audio denoising techniques: Spectral Gating, Butterworth Filtering, Gaussian Filtering, and Wavelet Denoising. We utilize the signal-to-noise ratio of speech signals (SNR), Root Mean Square Error (RMSE), and Short-Time Objective Intelligibility (STOI) as statistical metrics to compare the effectiveness of the denoising techniques. After statistical comparison we establish that the Spectral Gating denoising technique is superior to the other techniques in denoising and producing an intelligible audio signal.

Introduction

The increasing reliance on audio conferencing for education, business operations, and other aspects of life has reinforced the need for clear communication. This has created a demand for Audio Denoising techniques and how to effectively use them for the various types of noises.

This research presents an exploratory analysis of four audio denoising methods: Spectral Gating, Butterworth Filtering, Gaussian Filtering, and Wavelet Denoising.

To evaluate the effectiveness of these methods, we utilize three metrics: Signal-to-Noise Ratio (SNR), Root Mean Square Error (RMSE), and Short-Time Objective Intelligibility (STOI). These metrics provide insight into both the presence of noise in a signal and the perceived intelligibility of speech after denoising.[1]

Description of Data

We sourced our speech samples from librispeech[2] and our noise samples from a Kaggle repository[3]. After merging and denoising the audio we had a total of 120 denoised audio samples for evaluation.

Methodology

Dataset: We constructed 30 noisy speech clips by mixing six clean speech recordings (3 male, 3 female, varied inflections) with five background noises: *Keyboard Typing*, *Sprinkler*, *Fire Crackling*, *Chainsaw*, and *Water Droplets*[3]. These noises range from stationary (e.g., sprinkler) to highly nonstationary (e.g., keyboard).

Denoising Techniques: Each noisy sample was processed using four denoising methods:

- **Spectral Gating:** Removes noise in the frequency domain using a spectral mask based on an estimated noise profile.
- **Butterworth Filtering:** Applies a low-pass filter with a 4000 Hz cutoff to attenuate high-frequency noise.
- **Gaussian Filtering:** Smooths the signal using a 1D Gaussian kernel to suppress noise in the time domain.
- **Wavelet Denoising:** Uses discrete wavelet transform (DWT) and soft thresholding to reduce high-frequency noise while preserving speech features.

After denoising the 30 noisy audio samples with our 4 techniques, we computed our metrics (SNR, RMSE, and STOI). After organizing our metric data we applied statistical tests (Kruskal Wallis Test[4], Bonferroni Pair-Wise Test, Median Ranking)

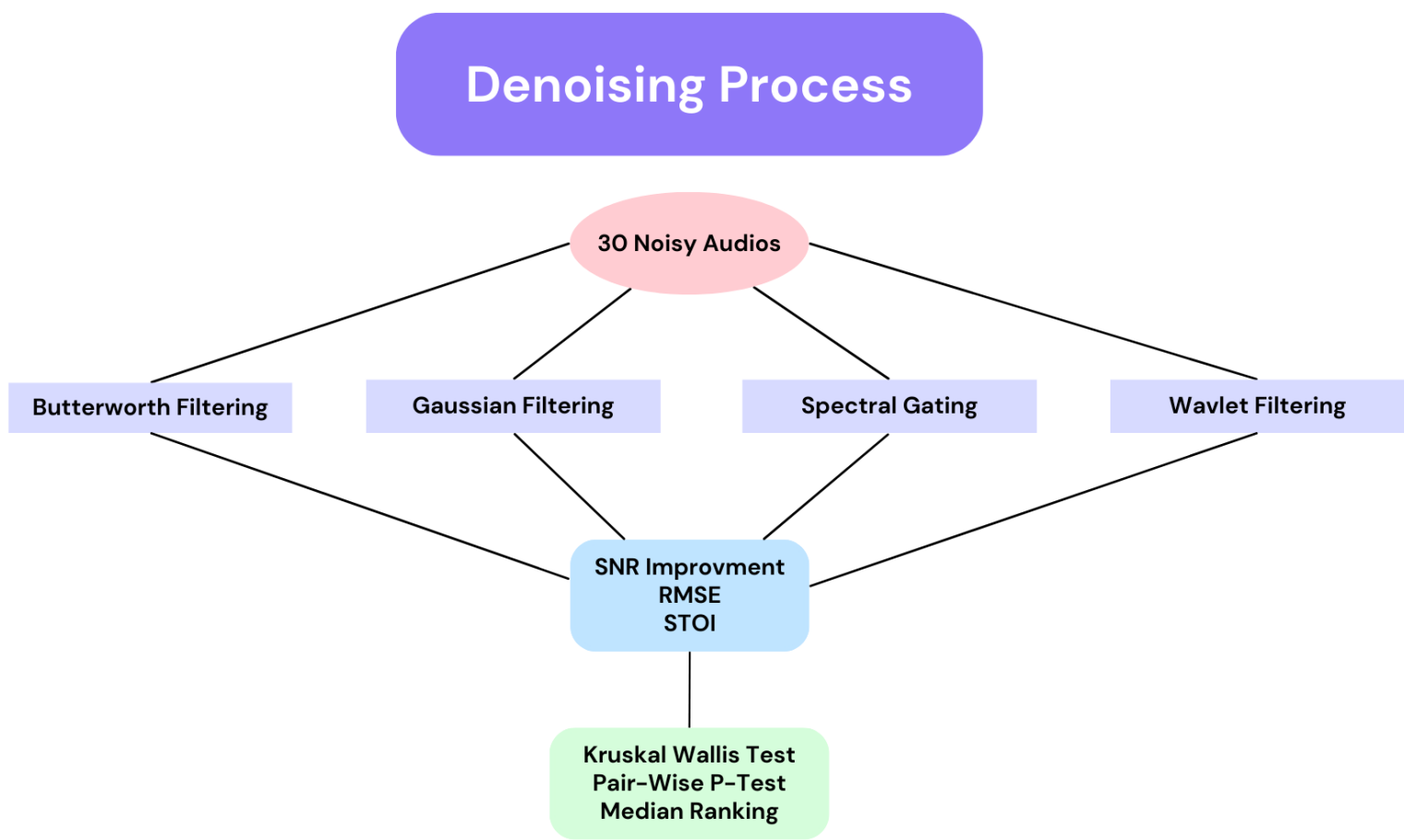


Figure 1: Denoising Process Flowchart.

Evaluation Metrics:

- **SNR Improvement (dB):** Measures the difference in signal-to-noise ratio after denoising:

$$SNR_{Improvement} = SNR_{denoised} - SNR_{noisy}$$

Higher values indicate more effective noise suppression. Where SNR is computed in the following manner where P is the power of a signal:

$$SNR = 10 \cdot \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

- **RMSE:** Computes the average error between the clean and denoised signals. Lower values indicate higher fidelity to the original signal:

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (x(t) - \hat{x}(t))^2}$$

Where $x(t)$ is the original clean signal and $\hat{x}(t)$ is the denoised signal.

- **STOI:** Predicts speech intelligibility on a scale from 0 (poor) to 1 (perfect) by comparing short-time envelopes of clean and denoised signals.

All metrics were computed for each of the 120 samples to compare denoising performance.

Explanatory Data Analysis

Normality Analysis

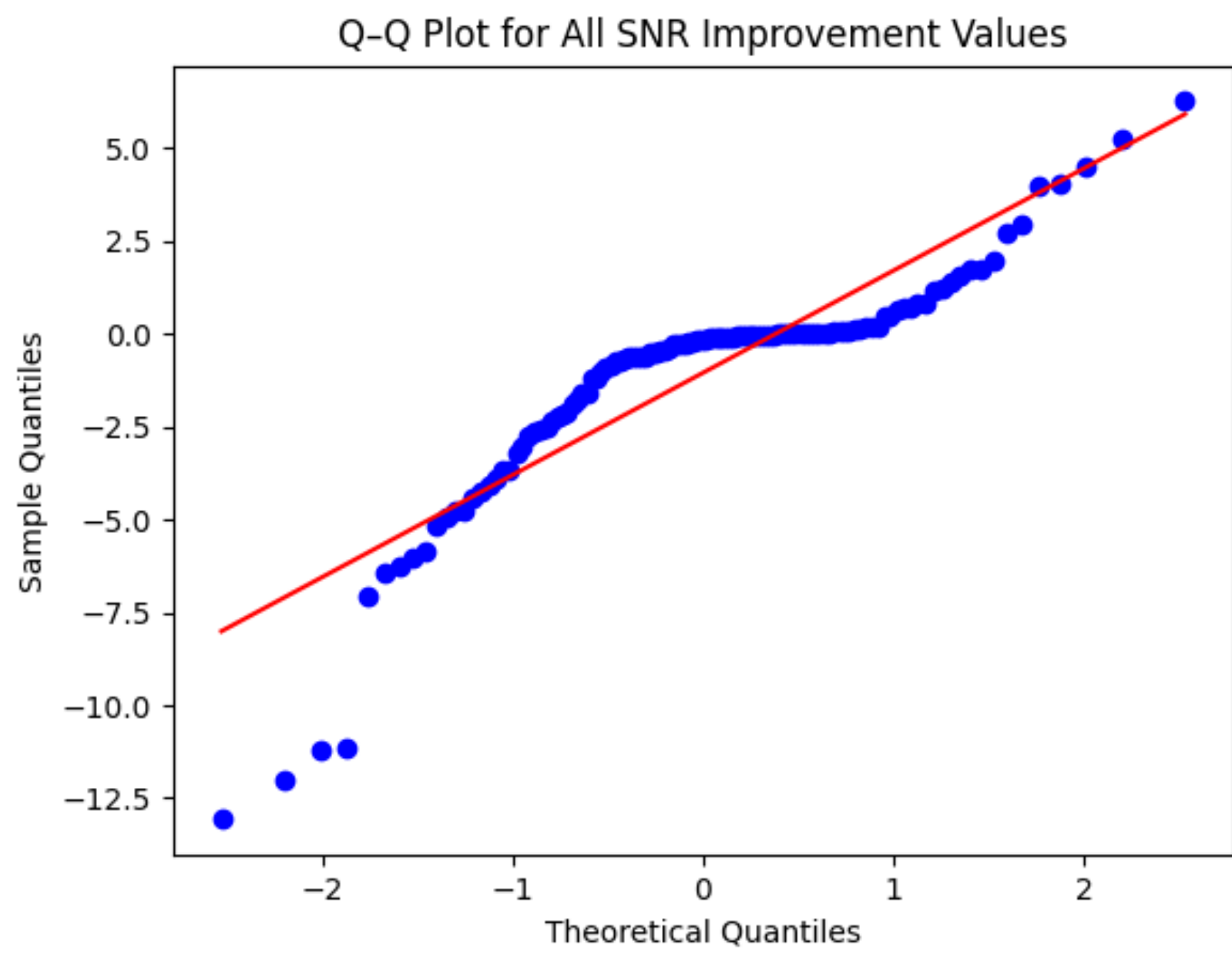


Figure 2: Q-Q Plot for SNR Improvement

The Shapiro-Wilk test ($\alpha = 0.05$) showed the SNR data were not normally distributed (statistic = 0.9072, $p \approx 0.00$). We therefore used the Kruskal-Wallis test to assess median SNR differences across methods, followed by Bonferroni-corrected pairwise comparisons to identify specific differences.

SNR Correlation Matrix (Kruskal-Wallis H-statistic = 13.805, P-value = 0.0032)				
	Butterworth	Gaussian	Spectral Gating	Wavelet
Butterworth	1.000000	0.002794	1.000000	0.321715
Gaussian	0.002794	1.000000	0.042304	0.698630
Spectral Gating	1.000000	0.042304	1.000000	1.000000
Wavelet	0.321715	0.698630	1.000000	1.000000

Table 1: Correlation matrix for SNR improvement between denoising methods.

Statistical testing indicates significant differences in SNR improvement across methods. Gaussian stands out as the most inconsistent. Spectral Gating and Butterworth offer the most stable SNR improvements.

RMSE Correlation Matrix (Kruskal-Wallis H-statistic = 3.154, P-value = 0.3685)				
	Butterworth	Gaussian	Spectral Gating	Wavelet
Butterworth	1.000000	0.874141	1.000000	1.000000
Gaussian	0.874141	1.000000	0.886540	0.886540
Spectral Gating	1.000000	0.886540	1.000000	1.000000
Wavelet	1.000000	0.886540	1.000000	1.000000

Table 2: Correlation matrix for RMSE between denoising methods.

No significant difference in RMSE across methods. RMSE patterns are consistent across methods, with minimal variation in signal error.

Median Metric Values by Denoising Method			
Method	Median SNR	Median RMSE	Median STOI
Butterworth	0.001550	0.028205	0.71575
Gaussian	-1.685937	0.043183	0.73760
Spectral Gating	-0.013920	0.027203	0.80160
Wavelet	-0.440027	0.027203	0.75570

Table 3: Median metric values for each denoising method.

Spectral Gating provided the best overall balance between clarity and fidelity, outperforming others in RMSE and STOI.

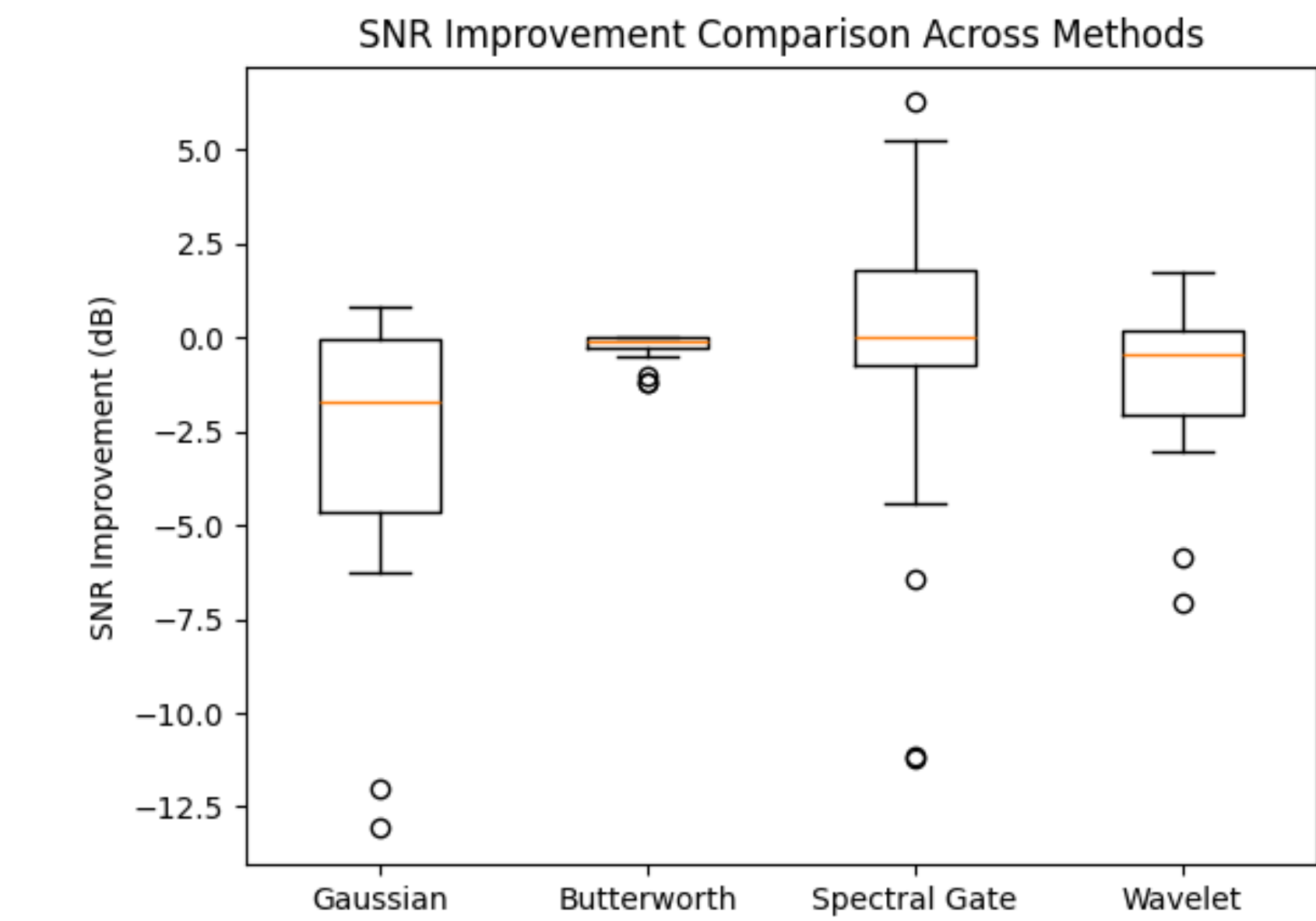


Figure 3: Boxplot of SNR Improvement Values

Results

Based on our evaluation, Spectral Gating emerged as the most effective denoising method overall. While each technique demonstrated unique strengths, Spectral Gating consistently achieved the lowest RMSE and highest STOI scores, indicating strong performance in both signal fidelity and intelligibility. Although the Butterworth Filter showed a slightly higher median SNR improvement and a very consistent SNR improvement, the difference was statistically insignificant when compared to Spectral Gating. Gaussian filtering performed inconsistently across all measures. These results suggest that Spectral Gating offers a well-balanced tradeoff between noise reduction and speech clarity across varied noise conditions.

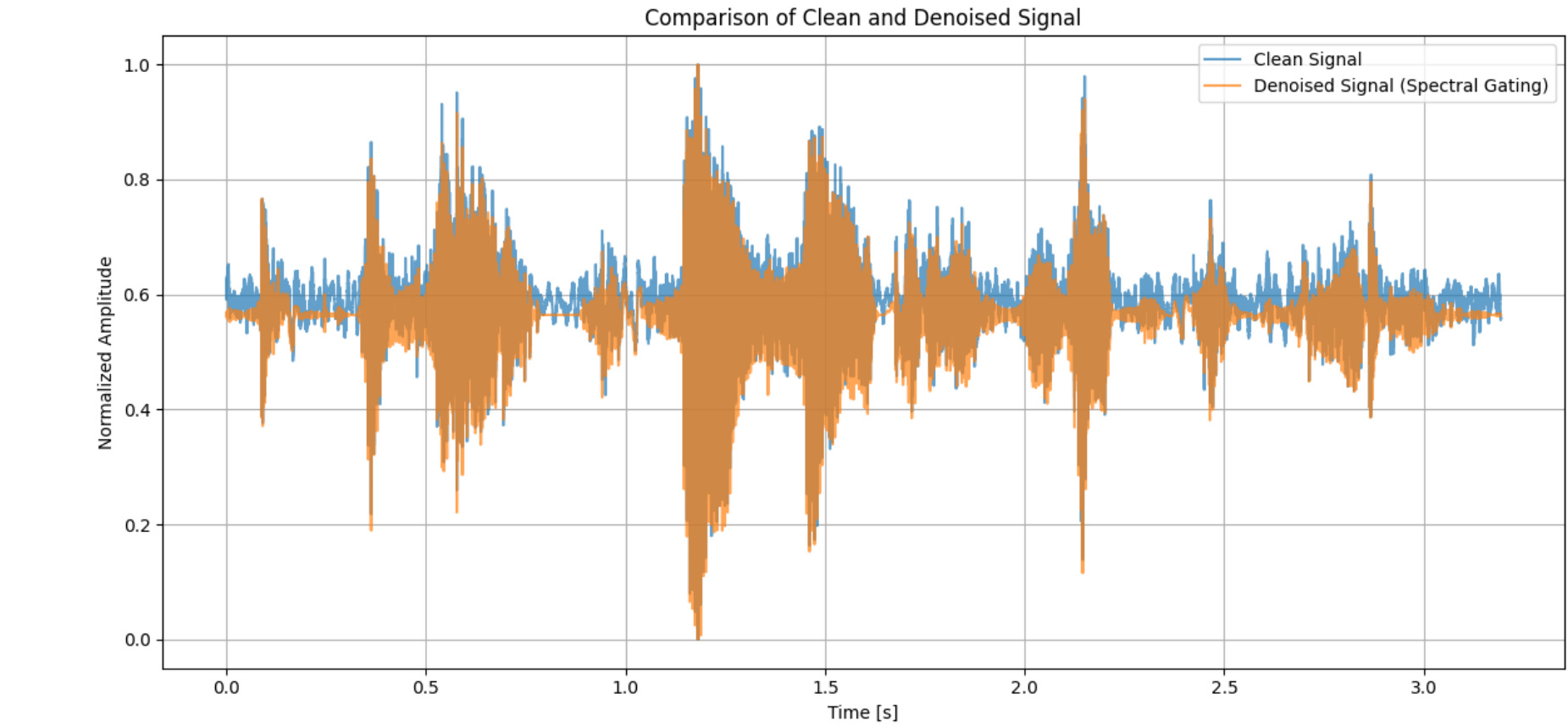


Figure 4: Spectral Gating Denoised vs Clean Signal Waveform

Conclusions

This study evaluated four denoising methods, Spectral Gating, Butterworth, Gaussian, and Wavelet, using SNR, RMSE, and STOI as performance metrics. Spectral Gating consistently delivered the best results, with the lowest RMSE and highest STOI, indicating strong noise reduction and speech intelligibility. While Butterworth achieved the highest median SNR, the improvement over Spectral Gating was minor. Overall, Spectral Gating offered the most effective trade-off across varied noise conditions. Our Future work will expand to include more diverse noise types and investigate deep learning-based techniques.

References

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